Date:	31 January 2025	**Maximum Marks:**	2 Marks
Team ID:		**Project Name:**	E-commerce application

IDEATION PHASE

Define the Problem Statements

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1. INTRODUCTION & CONTEXTUAL FRAMEWORK

In the architectural lifecycle of modern software engineering, particularly within modern web application paradigms like the MongoDB, Express.js, React, and Node.js (MERN) stack, systematic ideation establishes the core product trajectory. This phase shifts the design narrative away from purely programmatic constraints toward user-centric optimization. Establishing a thorough, empathetic framework for user interactions ensures that structural databases, server-side configurations, and client interfaces map explicitly to verified customer challenges.

To establish full market alignment for our proposed full-stack E-Commerce platform, this document uses the structured Customer Problem Statement Template. By decomposing user requirements into explicit statements of identity, intent, barriers, root causes, and subsequent emotional impact, the development team can construct robust technical solutions targeted precisely at widespread user pain points.

2. PROBLEM STATEMENT METHODOLOGY

The analytical framework utilized herein systematically breaks down user behavior into five core psychological and operational benchmarks:

- **I am (Customer Identity):** Defines the archetypal user, detailing 3–4 definitive behavioral or demographic attributes relevant to platform operational access.
- **I'm trying to (User Objective):** Pinpoints the programmatic or operational goal that the target customer actively seeks to execute or resolve within the ecosystem.
- **But (System/Process Barrier):** Identifies the modern friction point, system lag, architectural omission, or operational hurdle currently restricting completion.
- **Because (Root Cause Analysis):** Pinpoints the underlying structural, technical, or systemic deficiency driving the identified transactional friction.
- **Which makes me feel (Affective Impact):** Tracks the psychological and emotional impact experienced by the target demographic, defining indicators like user retention risks or platform abandonment metrics.

3. EMPIRICAL CUSTOMER PROBLEM MATRIX

The matrix below articulates distinct, deeply analyzed user challenges identified within contemporary multi-vendor e-commerce application models. These entries directly govern upcoming architectural schemas, backend route logic, API optimization targets, and state management implementations.

PROBLEM STATEMENT	**I AM** (Customer)	**I'M TRYING TO** (User Objective)	**BUT** (System Barrier)	**BECAUSE** (Root Cause)	**WHICH MAKES ME FEEL** (Affective Impact)
PS-1	A busy, tech-savvy mobile shopper who prioritizes immediate checkout pipelines.	Securely complete an order purchase using a mobile browser during a brief daily work commute.	The checkout interface experiences structural layout shifts and processing lag times exceeding 8 seconds.	The current monolithic platform architecture lacks localized client state caching and relies on unoptimized, uncompressed asset bundles.	Frustrated, highly impatient, and statistically primed to abandon the digital shopping cart.
PS-2	A value-conscious consumer tracking strict monthly household utility budgets.	Compare historical product pricing variations and track localized discount codes across multiple merchant categories.	The platform displays fragmented, inconsistent pricing models and fails to apply verified promo structures uniformly.	The relational database contains highly redundant data layers and lacks atomic transactional logic for synchronized API updates.	Skeptical, financially insecure regarding platform integrity, and discouraged from returning.